SOFTWARE ENGINEERING – LEVEL 3 MINI-PROJECT

DELIVERABLE 1 – SYNOPSIS / PROJECT PROPOSAL

Online Exam Portal – Web-Based Examination, Question Bank and Result Management System

1.1 Project Title and Group

Field	Details
Team Members	Samarth Kotian – PES1UG24CS406 / Saikavin S – PES1UG24CS402 / Samrudh C Patil – PES1UG24CS417 / Satyam Goyal – PES1UG24CS424
Project Title	Online Exam Portal – Web-Based Examination, Question Bank and Result Management System
Group / Team Number	Team 13 (G-13)
Project ID	OEP-2026-13
Course	UE24CS341A – Software Engineering, V Semester
Development Methodology	Agile (Scrum, 6 sprints)
Technology Stack	JavaScript – MERN Stack (MongoDB, Express.js, React.js, Node.js)
Complexity	Hard
Customer / Stakeholder	Department Examination Cell / Course Faculty

1.2 Team Profile and Product Ownership

Member	SRN	Owned Functional Feature	Secondary / Qualitative Responsibility
Samarth Kotian	PES1UG24CS406	Question Bank & Exam Evaluation — question creation/editing, categorization, randomization, objective evaluation, scoring and result generation.	Database architecture and data consistency.
Saikavin S	PES1UG24CS402	Authentication & Administration — registration/login, JWT authentication, RBAC, administrator functions and user/system management.	Security validation and access-control testing.
Samrudh C Patil	PES1UG24CS417	Exam Management & Attempt Processing — exam creation/configuration, scheduling, attempt lifecycle, auto-save and automatic submission.	REST API reliability, validation and error handling.
Satyam Goyal	PES1UG24CS424	Student Examination & Performance — examination interface, question navigation, result viewing, history and performance dashboard.	Usability and responsive student experience.

1.3 Proposed Project Description

The Online Exam Portal is a web-based examination management system designed for educational institutions. It digitizes the examination workflow from question-bank preparation and exam scheduling to timed student attempts, automatic evaluation, result generation and performance reporting. The system provides a centralized, secure and manageable platform for students and examination administrators/faculty.

1.4 Problem Statement

Examinations conducted through manual or ad-hoc digital processes depend heavily on physical question paper handling, in-person invigilation and manual, error-prone evaluation. Students have little visibility into their attempt history or performance trends, and faculty/administrators lack a centralized system to author question banks, schedule exams, control access by role, and generate consistent, tamper-resistant results. The Online Exam Portal digitizes the end-to-end examination lifecycle — from question bank creation to timed student attempts, automatic evaluation and result analytics — while keeping the workflow simple for both students and administrators.

1.5 Target Users

User Type	Primary Need	Major Activities
Student	Attempt scheduled examinations and view outcomes.	Login, view available exams, attempt timed exams, navigate questions, submit answers, view results/history and performance.
Administrator / Faculty	Manage examinations and examination content.	Manage users, create/manage question banks, configure and schedule exams, monitor exam operations and view reports.

1.6 Objectives

- Provide secure, role-based access for students and administrators using JWT-based authentication.
- Allow administrators to build and maintain a reusable, categorized question bank.
- Allow administrators to create, configure and schedule exams (duration, question set, availability window).
- Randomize question order/selection per student attempt to reduce malpractice.
- Allow students to attempt exams within a fixed time limit, with clear navigation between questions.
- Automatically evaluate objective questions and generate results immediately on submission.
- Provide students with a performance dashboard and administrators with operational/exam analytics.
- Develop the product incrementally using Agile practices on the MERN stack.

1.7 Scope

In Scope	Out of Scope
• Student and administrator registration/login and role-based access.	• AI/webcam-based online proctoring.
• Question bank creation, editing, categorization and disabling.	• Automated evaluation of descriptive/subjective answers.
• Exam creation, configuration and scheduling.	• Integration with external university examination/ERP systems.
• Question randomization/shuffling for attempts.	• Native Android/iOS mobile application.
• Timed online examination with navigation, marked-for-review and auto-save.	• Live audio/video invigilation and surveillance.
• Automatic submission on time expiry.	
• Automatic evaluation and scoring of objective questions.	
• Result generation, student history/performance views and administrator reports.	

1.8 Functional Features

1. Student and administrator registration and login (JWT authentication).
2. Role-based authorization and access control (Student vs. Admin).
3. Admin question bank management — add, edit, categorize and disable questions.
4. Admin exam creation and scheduling — assign questions, set duration and availability window.
5. Question randomization/shuffling per attempt.
6. Student exam attempt interface with question navigation (answered / unanswered / marked for review).
7. Auto-save of responses and automatic submission when time expires.

8. Automatic evaluation and scoring engine for objective questions.
9. Result generation and student performance/history view.
10. Admin reporting — exam-wise and student-wise operational reports.

1.9 Proposed Technology / Tool Stack

Area	Proposed Technology / Tool
Frontend	React.js
Backend	Node.js + Express.js
Database	MongoDB
Authentication	JWT
Programming Language	JavaScript
Version Control	Git + GitHub
Agile / Issue Tracking	Jira or GitHub Projects
API Testing	Postman
Automated Testing	Jest / appropriate JavaScript testing tools
CI/CD	Jenkins
Containerization	Docker
Static Analysis	SonarQube or similar

1.10 Agile Methodology Justification

The project will follow Agile Scrum because the system consists of multiple modules that can be developed and validated incrementally. Six sprints will progressively deliver authentication, question-bank management, exam management and attempts, evaluation/results, analytics/security hardening, and final system validation/release. Regular sprint reviews, testing and integration will help the team identify issues early and maintain continuous progress toward a working product.

1.11 Short-Term Plan

Iteration	Planned Output	Acceptance Evidence
Sprint 1	Project setup, requirements baseline, JWT authentication (register/login)	User stories + Git repository + working login/register
Sprint 2	Question bank module (CRUD) and admin exam creation	Demo + unit tests
Sprint 3	Student exam attempt engine — randomization, navigation, auto-save	Integration tests + PRs
Sprint 4	Evaluation & scoring engine, result generation	System demo + test evidence
Sprint 5	Student performance dashboard, admin analytics, security hardening	Validation + coverage report
Sprint 6	System validation, CI/CD, final release	Final report + deployment/demo